iNaturalist and EDDMapS provide complementary data for monitoring non-native herpetofauna in Florida

Fort Lauderdale Research and Education Center[1], Department of Wildlife Ecology and Conservation[2], University of Florida

L. Gallivan[2], M. Zuliani[1], *B.M. Mason[1], & C.T. Callaghan[1]*

Background

- Florida is a hotspot for non-native herpetofauna introductions, with 1 in 3 becoming established
- Citizen science platforms act as a tool to supplement biodiversity data monitoring
- Platform strengths and biases are not well quantified or compared to one another

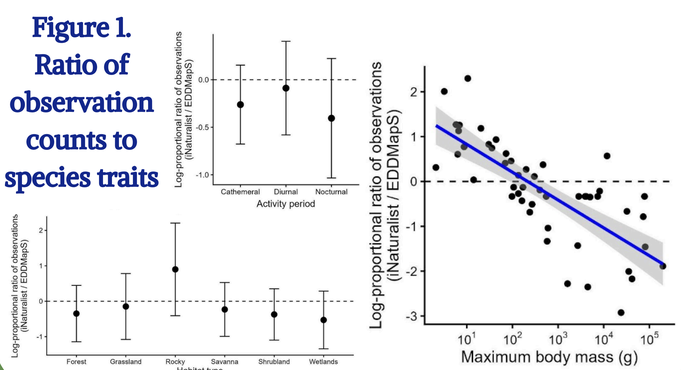
KEY OBJECTIVE:
Compare non-native herpetofauna records from iNaturalist, a more general and widespread platform, and EDDMapS, a more targeted platform from 2014-2024

Methods

Compared records across 3 components:

1. Data availability & species traits using generalized linear models (GLMs)
2. Urbanization bias
3. Spatial distribution & range size using minimum convex polygons (MCPs)

Figure 1. Ratio of observation counts to species traits



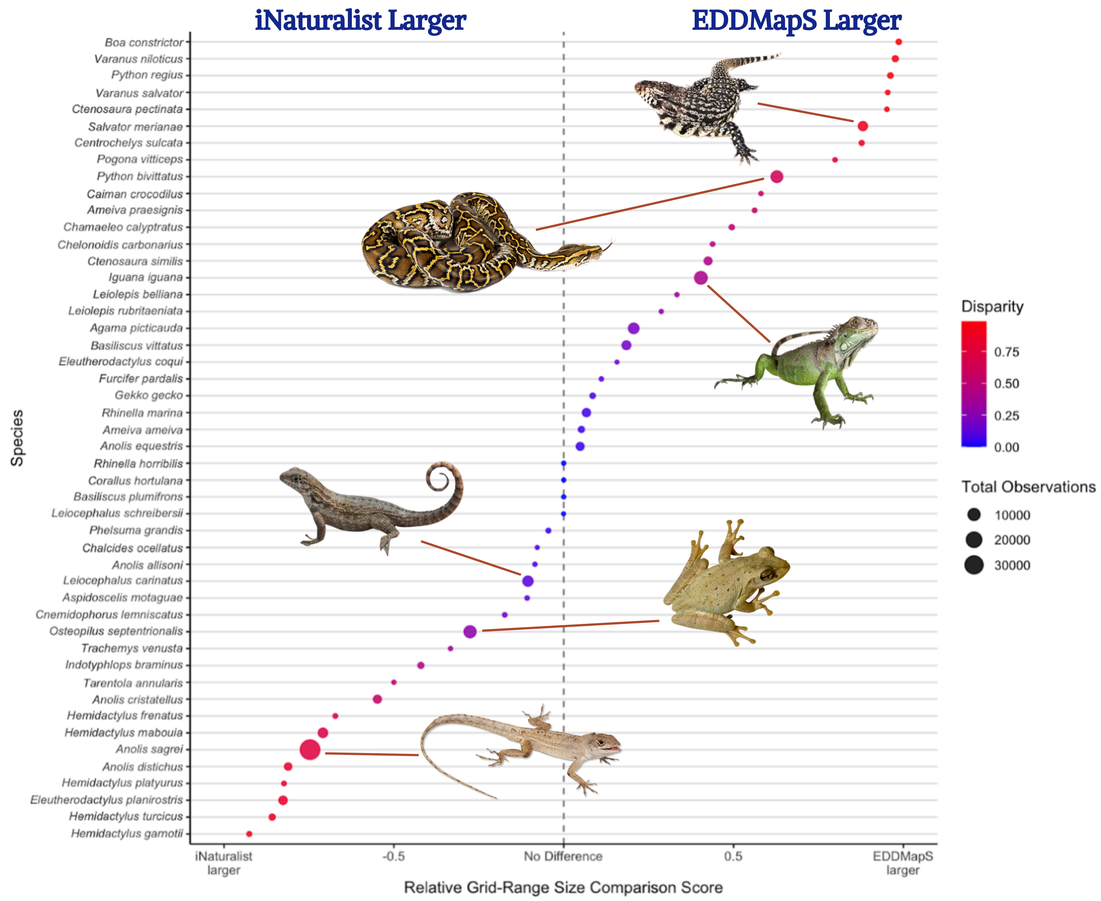


Figure 2. Comparison of grid-based range analysis between EDDMapS and iNaturalist records for non-native herpetofauna in Florida

Key Results

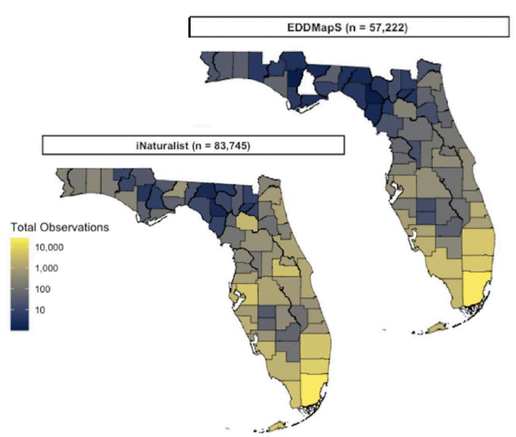


Figure 3. Spatial distribution of total observations per county by platform

- iNaturalist had more total observations
- 130 total species: 82 species on iNaturalist, 101 on EDDMapS
- Sampling effort is a major driver of spatial extent, range size
- After controlling for sampling effort, EDDMapS records covered a significantly larger grid-cell area than iNaturalist

Discussion

These findings:

- Suggest that each platform serves a role
  - **iNaturalist** → high volume, opportunistic sampling by general public
    - small bodied species
    - clustered urban hotspots
  - **EDDMapS** → targeted, professional effort
    - large-bodied species
    - broader spatial coverage
- Crucial to leverage both datasets to enhance early detection and mitigation strategies of herpetofaunal invasions

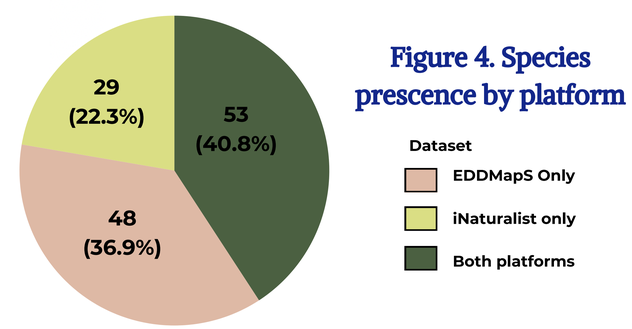


Figure 4. Species prescence by platform

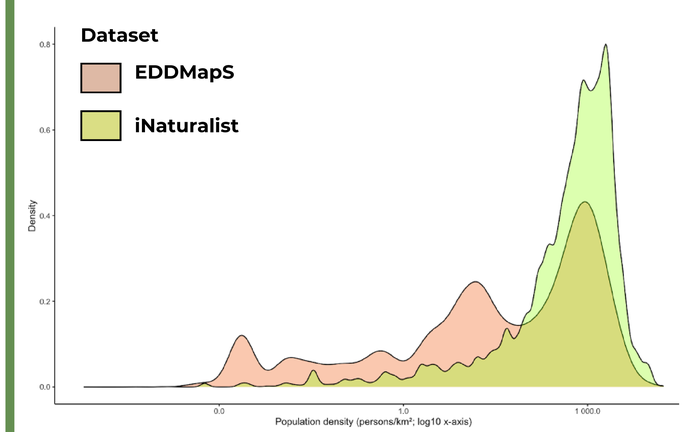


Figure 5. Distribution of observations along a log-transformed urbanization gradient